

Foundation (Jalapeno)

Q1. If $f(x) = x^2$ and $g(x) = 2x$, what is $(f + g)(x)$?

- (A) $x^2 + 2x$
- (B) $x^2 - 2x$
- (C) $2x^2$
- (D) $x^2 + x$
- (E) $x^2 - x$

Q2. Find the domain of $f(x) = \frac{1}{x}$ and $g(x) = \frac{1}{x-2}$. What is the domain of $(f \cdot g)(x)$?

- (A) $(-\infty, 0) \cup (0, 2) \cup (2, \infty)$
- (B) $(-\infty, 0) \cup (0, \infty)$
- (C) $(-\infty, 2) \cup (2, \infty)$
- (D) $(-\infty, 0) \cup (2, \infty)$
- (E) $(-\infty, 0)$

Q3. Given $f(x) = 2x - 1$ and $g(x) = x + 3$, what is $(f - g)(x)$?

- (A) $x - 4$
- (B) $x + 2$
- (C) $x - 2$
- (D) $x + 1$
- (E) $x - 5$

Q4. If $f(x) = x^2 + 1$ and $g(x) = x - 2$, what is $(f \div g)(x)$?

- (A) $\frac{x^2+1}{x-2}$
- (B) $\frac{x^2-1}{x-2}$
- (C) $\frac{x^2+1}{x+2}$
- (D) $\frac{x+2}{x^2+1}$
- (E) $\frac{x^2-2}{x+2}$

Q5. The orbital velocity v of a planet can be modeled by

$v(x) = \frac{2x}{x+1}$ and the escape velocity e by $e(x) = x + 2$. What is $(v + e)(x)$?

- (A) $\frac{2x}{x+1} + x + 2$
- (B) $\frac{3x}{x+1} + 2$
- (C) $\frac{x+2}{2x}$
- (D) $x^2 + 1$
- (E) $\frac{4x}{x-1}$

Q6. The distance d traveled by a spacecraft can be represented by $d(x) = 2x^2$ and the fuel consumption f by $f(x) = x + 1$. What is $(d - f)(x)$?

- (A) $2x^2 - x - 1$
- (B) $2x^2 + x - 1$
- (C) $2x^2 - x + 2$
- (D) $2x^2 + x + 1$
- (E) $2x^2 - 2x - 1$

Q7. If $f(x) = x^2 - 1$ and $g(x) = 2x + 3$, what is $(f \cdot g)(x)$?

- (A) $2x^3 + 3x^2 - 2x - 3$
- (B) $2x^3 - 3x^2 - 2x + 3$
- (C) $2x^3 + x^2 - 5x - 3$
- (D) $2x^2 + 3x - 2$
- (E) $x^3 + 4x^2 - 3x - 1$

Q8. Given $f(x) = \frac{2}{x}$ and $g(x) = x - 1$, what is the domain of $(f \div g)(x)$?

- (A) $(-\infty, 0) \cup (0, 1) \cup (1, \infty)$
- (B) $(-\infty, 0) \cup (1, \infty)$
- (C) $(-\infty, 1) \cup (1, \infty)$
- (D) $(-\infty, 0) \cup (0, 1)$
- (E) $(-\infty, 0)$

Open Ended Questions (FRQ)

Q9. If $f(x) = x^2 + 2x$ and $g(x) = 3x - 1$, find $(f + g)(x)$ and simplify. Then, determine the domain of $(f + g)(x)$. Show all work.

Q10. A lab experiment involves two functions: $f(x) = \frac{x}{x+1}$, representing the concentration of a chemical, and $g(x) = x^2 - 2$, representing the volume of the container. Find $(f \cdot g)(x)$ and simplify. What is the practical meaning of $(f \cdot g)(x)$ in this context? Show all work and explain your answer.

Chef's Tips

- Emphasize simplification of resulting expressions after operations.
- Focus on understanding the concept of function operations and their domains.
- Use the context of space and science to make word problems unique and engaging.